Haotian Liu

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Selected Publications & Manuscripts

ACCEPTED

1. PilotBench: A Benchmark for General Aviation Agents

Yalun Wu*, Haotian Liu*, Zhoujun Li, Boyang Wang

IEEE International Joint Conference on Neural Networks (IJCNN 2026). *arXiv:2604.08987*

A flight-agent benchmark built from 708 real-world trajectories and 34-channel telemetry. Evaluation across 41 models identifies a precision–controllability trade-off and a dynamic-complexity gap in high-workload flight phases. Pilot-Score combines regression accuracy with instruction following and constraint compliance. Traditional forecasters reach MAE 7.01, while LLMs achieve 86–89% instruction following at 11–14 MAE.

2. FLY-EVAL++: An Evidence-Driven Evaluation Protocol for Flight Prediction with Large Language Models

Yalun Wu, Junfeng Fang, Jiawei Wang, Haotian Liu, Qijun Yang, Minghan Yang, Hongcheng Guo, Zhoujun Li, Boyang Wang

Conference on Language Modeling (CoLM 2026). *arXiv:2609.04021*

An evidence-driven protocol for structured validity, physical feasibility, and constraint compliance in flight prediction. Across 66 LLMs, predictive quality and constraint compliance emerge as distinct capability dimensions, with a 28.2-point constraint gap at comparable predictive quality.

UNDER REVIEW

3. Beyond Answer Voting: Fine-Grained Correction of LLM Reasoning through Graph Alignment across Sampled Traces

Haotian Liu, Yingjian Chen, Xiaohang Xu, Lang Huang, Yusuke Iwasawa, Yutaka Matsuo, Irene Li

ACL Rolling Review (ARR) – Under review. First author. *arXiv:2610.18606*

Introduces DDRG (Disagreement-Driven Verification Guided by Reasoning Graphs), which builds source-grounded claim graphs from a frozen bank of sampled traces, sends aligned claim-level conflicts to an answer-blind verifier, and triggers one focused repair only when a refutation reaches the majority answer's supporting path; otherwise it returns the exact self-consistency vote. Across five benchmarks, zero-shot DDRG improves macro-average accuracy over direct answering by 8.65 and 15.83 points for Qwen2.5-7B and Llama3.1-8B.

4. LFUSE: Learned Evidence Fusion for Noisy Spoken and Image-Grounded Product Search

Haotian Liu, Xiaowen Liu, Yunhan Wang, Yalun Wu, Minghan Yang, Xinhao Yu, Boyang Wang

ICASSP 2027 – Under review. First author.

A learned evidence-fusion model that reranks the union of BM25 and dense candidates using rank, character, pinyin, and query-agreement features, with noise-aware training for corrupted and ASR queries. It improves MRR@10 over tuned convex fusion by 7.8% on average across three human-judged Chinese Multi-CPR domains and lowers fusion-stage latency by 80–100× versus LLM-based adaptive fusion.

5. Before the Agent Acts: Understanding, Task Accounts, and Implementation in Pre-Coding Review

Haotian Liu, Mingyue Zhang, Nianyu Li, Eunsuk Kang, Jialong Li

Empirical Software Engineering (EMSE) – Under review. First author.

Introduces Collaborative Evidence-Constrained Task Interpretation (CETI), a repository-grounded pre-coding review workflow with editable, source-linked task Records. A 24-participant study over 72 cases reports a +17.2 percentage-point adjusted Records-2 versus FreeChat-2 contrast in objective understanding; a linked audit traces 102 request occurrences through 93 families and 504 coding attempts across 10 tasks and 3 models. Exact criteria cover 20 of 24 core families and 4 of 25 boundary families.

MANUSCRIPTS

6. Long-Horizon Memory Governance for LLM Agents (LHMG)

Haotian Liu (first author)

Manuscript in preparation; not yet submitted.

A modular governance layer that treats agent memory as auditable state, combining append-only revision records, active-state semantics, No-Silent-Overwrite updates, risk-gated conditioning, and governed forgetting. Controlled diagnostics show a 13.3-point GPT-4o gain in cross-turn decision consistency over flat memory.

* Equal contribution.